

Reject inference

When a lender builds a credit scorecard, it only knows the repayment outcome of **accepted** applicants. Rejected applicants never receive a loan, so their future default or non-default outcome is unknown. This creates sample selection bias because the model is trained on a group that is different from the group it will be used on. Reject inference tries to estimate the likely outcomes of rejected applicants and include them in model development.

1) Hard cut-off

Hard cut-off is the simplest augmentation method. The lender scores rejected applicants, picks a probability threshold, and labels each reject as Good or Bad based on that threshold.

Example: if the cut-off is 40%, then an applicant with 8% PD is labeled Good, while an applicant with 65% PD is labeled Bad.

Pros

- Very easy to implement.
- Easy to explain to management and regulators.
- Cheap and fast to compute.
- Can work reasonably well when the model separates good and bad applicants clearly.

Cons

- Creates artificial certainty.
- Small PD differences can flip labels sharply.
- Errors in the first model flow into the next one.
- Can produce unstable results.

2) Fuzzy augmentation

Fuzzy augmentation does not force one label on each rejected applicant. Instead, it splits each reject into two weighted records, one Good and one Bad, using the predicted probability.

Example: if a reject has 30% PD, you create one Good record with weight 70% and one Bad record with weight 30%.

Pros

- Uses probability information better than hard cut-off.
- Avoids arbitrary thresholding.
- Reflects uncertainty more realistically.
- Usually smoother than hard labeling.

Cons

- Harder to explain.
- Increases dataset size.
- Depends heavily on the calibration of the original model.
- May not consistently beat the financed-only model.

3) Parceling

Parceling groups rejected applicants into score bands and assigns bad rates by band. The bad rate in each band is usually based on the accepted population and then adjusted upward for rejects.

Example: if the accepted population has a 20% bad rate in a band, and rejects are assumed to be twice as risky, the inferred reject bad rate becomes 40%. If 1,000 rejects fall in that band, about 400 may be labeled Bad and 600 Good.

Pros

- Common in traditional banking scorecard work.
- More realistic than a hard cut-off.
- Preserves score distribution.
- Allows business judgment.

Cons

- Depends on subjective assumptions.
- Results change a lot based on the inflation factor.
- Different analysts may get different models.
- Hard to validate empirically.

4) Proportional assignment

Proportional assignment labels rejects randomly according to an assumed bad rate. It does not use individual applicant scores to make the assignment.

Example: if the reject bad rate is assumed to be 30%, then out of 10,000 rejects, about 3,000 are labeled Bad and 7,000 Good at random.

Pros

- Very simple.
- Quick to implement.
- No complex scoring is needed.

Cons

- Ignores applicant characteristics.
- Adds a lot of noise.
- Weak predictive value.

5) Reweighting

Reweighting does not create labels for rejects. Instead, it changes the weights of accepted applicants so they better represent the full applicant pool.

Example: if low-income applicants are underrepresented because they were often rejected, similar accepted cases may get a higher training weight so the model is less biased.

Pros

- No artificial labels.
- Statistically cleaner than label-guessing methods.
- Easier to govern.
- Often preferred by model validators.

Cons

- Needs a good model of the acceptance process.
- Wrong weights can bias the model.

- Some weights may become too large.

6) Self-learning

Self-learning is a semi-supervised method. The model is trained on accepted applicants, scores the rejects, labels only the most confident rejects, then retrains and repeats.

Example: if one reject has 2% PD, it may be labeled Good; if another has 95% PD, it may be labeled Bad; the uncertain middle cases are left unlabeled for now.

Pros

- Uses unlabeled data effectively.
- Can perform better than simple augmentation.
- May improve discrimination.

Cons

- Mistakes can accumulate across iterations.
- Needs careful monitoring.
- More computationally intensive.

7) Deep generative or Bayesian methods

These methods use probabilistic models to jointly model accepted applicants, rejected applicants, and repayment outcomes. The outcome for each reject is treated as uncertain rather than forced into one label.

Example: instead of saying a reject is Good, the model may estimate a 82% chance of Good and use that probability directly in training.

Pros

- Can capture complex patterns.
- Handles non-linear relationships.
- Treats uncertainty more properly.
- Strong research potential.

Cons

- Hard to explain.

- More expensive to compute.
- Difficult to validate.
- Can be problematic for governance and regulation.

Practical comparison

Technique	Complexity	Explainability	Typical strength	Main weakness
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Hard cut-off	Low	High	Moderate	Artificial labels
Fuzzy augmentation	Medium	Medium	Moderate to high	Depends on calibration
Parceling	Medium	High	Moderate to high	Subjective assumptions
Proportional assignment	Low	High	Low	Ignores applicant differences
Reweighting	Medium	High	Moderate	Needs acceptance model
Self-learning	High	Medium	High	Error propagation
Deep generative / Bayesian	Very high	Low	High	Hard to govern

Ref:

- 1) <https://www.linkedin.com/pulse/reject-inference-overlooked-strategy-credit-scoring-bharath-t-s>
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- 3) <https://support.sas.com/resources/papers/proceedings09/305-2009.pdf>
- 4) <https://www.experian.com/blogs/insights/reject-inference/>
- 5) <https://pmc.ncbi.nlm.nih.gov/articles/PMC9041715/>
- 6) <https://www.sciencedirect.com/science/article/abs/pii/S0020025522004996>

Method	Main assumption	Main risk
Hard cut-off	The first model's predicted probability is good enough to turn rejects into Good/Bad labels.	A small PD change can flip the label, so the model may become unstable and carry forward early mistakes.
Fuzzy augmentation	Predicted probabilities are well calibrated enough to split each reject into weighted Good/Bad parts.	If the probabilities are miscalibrated, the weights are wrong and the training data becomes distorted.
Parceling	Score bands and inflation factors can reasonably approximate reject risk.	Results can change a lot depending on the analyst's assumptions, so the method may be subjective and hard to validate.
Proportional assignment	A portfolio-level bad rate is enough to assign reject outcomes.	It ignores individual applicant features, so it adds noise and weakens predictive power.
Reweighting	The acceptance process can be modeled well enough to correct sample bias.	If the acceptance model is wrong, the weights can be extreme or biased, which may hurt model stability.
Self-learning	The most confident pseudo-labels are mostly correct.	Early labeling mistakes can snowball through later iterations and damage performance.

Method	Main assumption	Main risk
Deep generative / Bayesian	The chosen probabilistic model can represent both accepted and rejected applicants well.	The model is harder to explain, validate, and govern, so it can create compliance and interpretability issues.

Ref:

- 1) Ehrhardt et al., *Reject inference methods in credit scoring*.
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<https://arxiv.org/pdf/1909.06108>
- 5) Mancisidor et al., *Deep Generative Models for Reject Inference in Credit Scoring*
<https://arxiv.org/pdf/1904.11376>